

Worksheet 3B: Quadratics — Factoring and Solving — Solutions

AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
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Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

Part A

1. Multiply to 6, add to 5; Numbers: 2 and 3; $(t + 2)(t + 3)$
2. $(p + 3)(p + 5)$
3. $(m + 4)(m + 5)$
4. $(r + 3)(r + 8)$
5. $(k + 2)(k + 5)$
6. $(n + 5)(n + 7)$

Part B

7. $(t - 2)(t - 3)$
8. $(p - 2)(p - 7)$
9. $(m - 3)(m - 5)$
10. $(r - 3)(r - 7)$
11. $(k - 4)(k - 7)$
12. $(n - 4)(n - 9)$
13. $(w - 3)^2$ — Perfect square!
14. $(s - 7)^2$ — Perfect square!

Part C

15. $(t + 5)(t - 2)$
16. $(p + 4)(p - 2)$
17. $(m - 5)(m + 3)$
18. $(r - 6)(r + 3)$

- 19. $(k + 7)(k - 3)$
- 20. $(n - 8)(n + 3)$
- 21. $(w + 6)(w - 5)$
- 22. $(s - 9)(s + 3)$
- 23. $(h + 11)(h - 4)$
- 24. $(u - 8)(u + 7)$

Part D

- 25. $(t + 5)(t - 5)$
- 26. $(p + 9)(p - 9)$
- 27. $(m + 10)(m - 10)$
- 28. $(2r + 3)(2r - 3)$
- 29. $(4k + 7)(4k - 7)$
- 30. $(5n + 1)(5n - 1)$
- 31. $(w + 5)^2$
- 32. $(s - 6)^2$
- 33. $(h + 8)^2$
- 34. $(2u + 3)^2$

Part E

- 35. $(2t + 3)(t + 1)$
- 36. $(3p + 4)(p + 2)$
- 37. $(2m + 1)(m + 4)$
- 38. $(5r + 3)(r + 2)$
- 39. $(3k - 1)(k - 2)$
- 40. $(2n - 1)(n - 5)$
- 41. $(2w + 3)(2w - 1)$
- 42. $(3s + 1)(2s - 3)$
- 43. $(3h + 2)(h - 4)$
- 44. $(2u + 5)(2u - 5)$
- 45. $(3v + 2)^2$
- 46. $2(q + 3)(q - 3)$

Part F

- 47. $2(t+2)(t+3)$
- 48. $5(p-1)(p-3)$
- 49. $3(m+3)(m-3)$
- 50. $4r(r+3)(r-1)$
- 51. $6(k-2)^2$
- 52. $2n(n-5)(n+1)$

Part G

- 53. $t = -3$ or $t = -4$
- 54. $p = 2$ or $p = 4$
- 55. $m = 3$ or $m = -5$
- 56. $r = 3$ or $r = -3$
- 57. $k = 5$ (one repeated solution)
- 58. $n = \frac{1}{2}$ or $n = -3$
- 59. $w = \frac{1}{3}$ or $w = 2$
- 60. $s = 0$ or $s = -8$
- 61. $h = 4$ or $h = -4$
- 62. $u = 5$ or $u = -2$
- 63. $v = -1$ or $v = -2$
- 64. $t = 6$ or $t = -2$

Part H

- 65. $t = 1$ or $t = -5$
- 66. $p = 9$ or $p = -1$
- 67. $m = -3$ or $m = -7$
- 68. $r = 5$ or $r = -3$
- 69. $k = -1$ or $k = -5$
- 70. $n = 6$ or $n = -2$
- 71. $w = 1$ or $w = -4$
- 72. $s = 6$ or $s = -1$
- 73. No real solutions (negative under square root)

74. No real solutions (negative under square root)

Part I

75. $t = 2$ or $t = 3$

76. $p = 2$ or $p = -5$

77. $m = -\frac{1}{2}$ or $m = -3$

78. $r = 1$ or $r = -\frac{1}{3}$

79. $k = 2 \pm \sqrt{3}$

80. $n = -1 \pm \sqrt{3}$

81. No real solutions (discriminant = -4)

82. No real solutions (discriminant = -31)

83. $h = 3$ (one repeated solution, discriminant = 0)

84. $u = -\frac{1}{2}$ (one repeated solution)

Part J

85. Positive; 2 solutions

86. Zero; 1 solution (repeated)

87. Negative; 0 real solutions

88. Positive (49); 2 solutions

89. Negative (-3); 0 real solutions

90. Positive (16); 2 solutions

91. Zero; 1 solution (repeated)

92. Negative (-7); 0 real solutions

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